Solids, liquids and gases

Remember that this work is about the particle model of solids, liquids and gases.

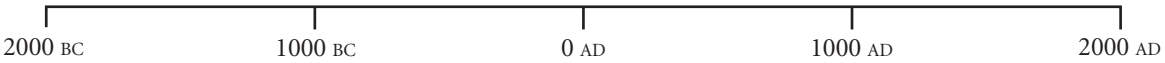
Two and a half thousand years ago in Greece, Parmenides had the idea that everything in the universe was made up of one kind of 'stuff'. Empedocles developed this theory, saying that everything was actually made up of one or a combination of four elements – earth, air, fire and water. Leusippus decided that this wasn't sensible and argued that everything was made up of tiny particles called atoms that could not be destroyed, Democritus developed this theory further, stating that there must be lots of different kinds of atoms, but all of them were tiny, round objects that couldn't change size or shape.

Some years later, Aristotle decided that the best explanation of 'what things are made of' was the theory of everything being made from four elements: earth, air, fire and water.

For the next 2000 years it was Aristotle's chosen theory that people read and believed.

It was only in the eighteenth century that a man called John Dalton did many experiments to try to work out what things were made of. When he looked at what he had found out he decided that everything must be made of tiny particles.

1 Copy the timeline and use the information above to put these theories in order on the timeline.



❶ Everything is made of Earth, Air, Fire and Water.

❷ Everything is made of tiny atoms.

❸ Everything is made of one kind of stuff.

❹ Everything is made of atoms which are tiny, round, hard objects that can't change shape. There are lots of different kinds of atoms.

❺ People believed that everything is made of Earth, Air, Fire and Water

❻ Experiments show that everything must be made of particles.

2 Look around your house for solids, liquids and gases. Complete a table like the one below:

Name of substance	Solid	Liquid	Gas

3 Use at the information you found in Question (2) to answer these questions

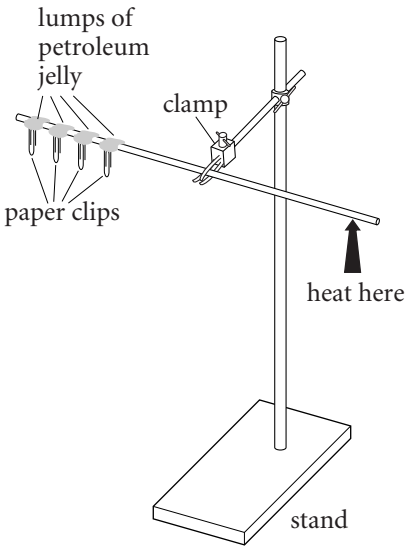
a Which are there most of: solids, liquids or gases?

b Pick ten of your substances from Question 2 (some solids, some liquids and some gases) and for each substance, say how its state helps it to be useful. For example a plate is a solid. This is useful because it keeps the same shape when you put food on it, keeping the food in one place.

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Heating solids

1 Copy these sentences and put them in order of what would happen using the apparatus above.



a The first paperclip falls off.

b Heat the end of the rod away from the paperclips.

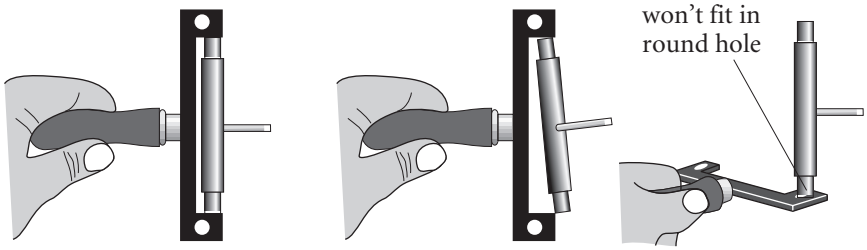
c The paperclip nearest the flame fell off first, then the next nearest and so on.

d The heat melts the wax holding the first paperclip onto the rod.

e The heat travels further along the rod.

f The paperclips fall off in order as the wax melts.

g The heat travels along the rod.



before heating after heating

2 When the bar and gauge apparatus shown above was cold, the bar would fit exactly between the two arms of the gauge, and the end of the bar would fit exactly in the round hole through the gauge.
When the bar was heated for about one minute, it would no longer fit between the arms at all, and the end of the bar would not fit in the round hole either.

a what has happened to the bar? Choose the best answer from:

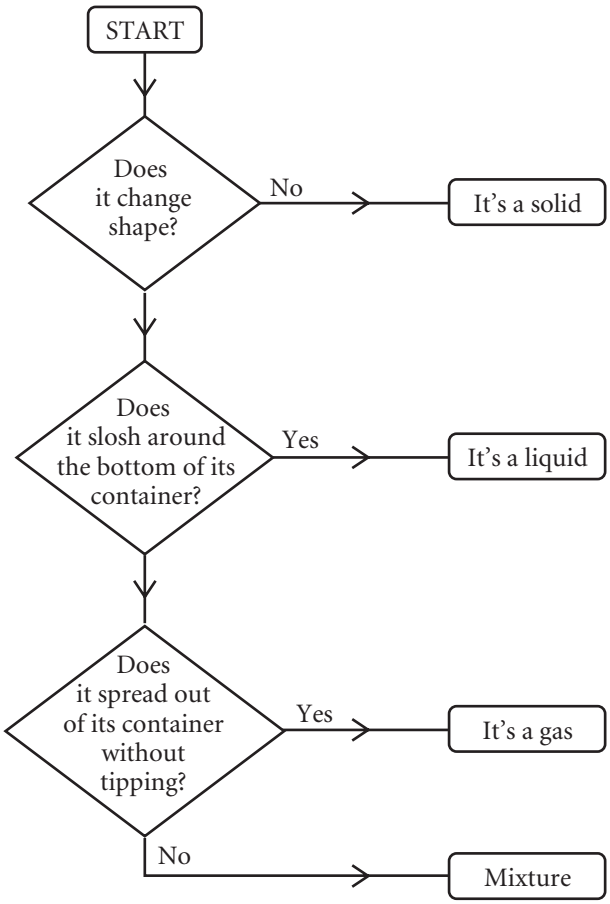
- the bar has got longer
- the bar has got shorter
- the bar has got thicker
- the bar has got thinner
- the bar has got longer and thicker
- the bar has got shorter and thinner
- the bar has not changed

b what has happened to the particles in the bar to make this happen? Choose the best answer from:

- the particles have changed colour
- the particles have got bigger
- the particles have got smaller
- the particles are closer together
- the particles are further apart

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Classifying solids, liquids and gases



1 Choose ten materials around your house. Try to get a range of different substances, and you might find it useful to choose some substances that have packets with a list of ingredients on them.

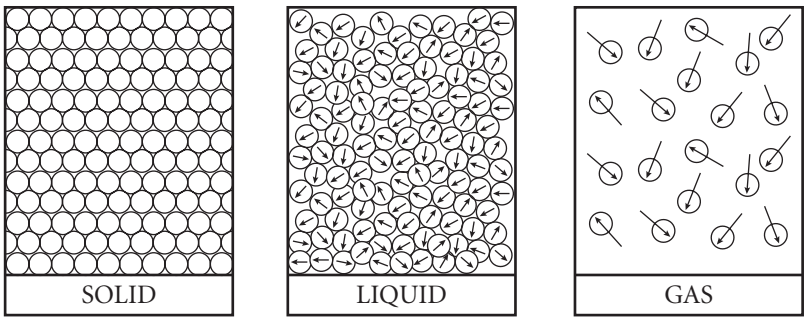
2 For each substance, use the key above to classify the substance as a solid, a liquid, a gas or a mixture.

3 For each substance, write a sentence explaining why you have classified the substance in that way. For example *''Tomato sauce always keeps the same size but does not keep the same shape very well, so it is a bit like a solid and a bit like a liquid. The list of ingredients shows that it is a mixture.''*

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Using the particle model

The particle model of solids, liquids and gases says that particles are arranged in different ways in solids, liquids and gases. The diagrams below show how the particles may be arranged in each of the three.



1 Which of the following statements are true, and which are false? You will need to read each statement very carefully.

a Particles are very small.

b Particles have a fixed size.

c Particles can always vibrate in all directions.

d Particles can always flow past each other.

e Particles are always fixed in place.

f Particles can always move wherever they want.

g Particles in a solid are free to move wherever they want.

h Particles in a gas are close together but can flow past each other.

i Particles in a liquid are fixed in place and can only vibrate.

j Particles in a solid are often arranged in a regular pattern.

2 Use the particle model to explain these statements:

a Solids stay the same size.

b Solids stay the same shape.

c Liquids can flow into new shapes.

d Liquids stay the same size.

e Gases can get bigger and spread out.

f Gases change their shape.

g Gases can be compressed.

h Liquids and solids can't be compressed

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Particles and pressure

1 Draw five diagrams to illustrate sentences **a** to **e**, which explain what happens when all the air is removed from a can.

- Show the air particles inside and outside the can.
- Draw a maximum of ten air particles inside the can.

a The can is full of air particles moving around. Some of them push against the inside walls of the can.

b Air particles push against the outside of the can.

c When the pump is turned on, there are fewer particles in the can.

d There are fewer air particles pushing against the inside
of the can wall, so the pressure in the can is lower.

e The higher air pressure outside the can squashes the can.

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Homework

H1

Dissolving in water

1 Look around your house for substances which might dissolve in water. Read the label (if there is one) to make sure there is no reason to keep the substance away from water.

- Check with an adult that it is OK to try dissolving the substance in water.
- Try to dissolve each substance you have found in a jar containing a little water.
- Draw and complete a table like the one below to record your results.

2 Jon doesn't believe that tap water is a mixture. Write a letter to him to explain how he could use evaporation to find out whether or not there is anything dissolved in tap water.

Name of substance	Soluble	Insoluble	Can't tell

Homework

H2

Mokhtar's muddle

1 Mokhtar's salt caravan was crossing the Sahara desert from Taroudannt to Tombouctou when one of his camels stumbled and spilled its load of salt into the sand. Mokhtar shovelled up all the salt and loaded it back on to the camel and continued on his way.

When he got to the market in Tombouctou he was dismayed to find that no-one would buy the salt because of the sand that was mixed with it. How could he purify the salt again?

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